Module 4: **Algebric Structure & Morphism** — Group Theory

Properties of Binary Operations

1. Closure Property
2. Associative Property
3. Identity Property
4. Inverse Property
5. Commutative Property.
6. Idempotent Property.

- Closure = Algebric Structure
- Closure + Associative = Semigroups
- Closure + Associative + Identity = Monoids
- Closure + Associative + Identity + Inverse = Groups
- Closure + Associative + Identity + Inverse + Commutative = Abelian Group.

① $\boxed{\text{Algebric Structure}}$
↓
Closure Properties. → Let A be non empty set. An operation $*$ is said to be closure on A if $\boxed{\forall a,b \in A \quad a*b \in A}$

$7+1=8 \in N$

$1-8=-7 \notin N$

eg.

N, +	Yes	Z, +	Yes	R, +	Yes
N, −	No.	Z, −	Yes	R, −	Yes
N, ×	Yes	Z, ×	Yes	R, ×	Yes
N, ÷	No	Z, ÷	No	R, ÷	No

② $\boxed{\text{Semi Groups}}$ Let $(A,*)$ be an algebric structure, where $*$ is a binary operation on A. $(A,*)$ is called semigroup if the conditions are satisfied.

- Closure Property.
- Associative Property.

$\boxed{a*(b*c)=(a*b)*c}$

$(3+2)+5 = 3+(2+5)$ ✓

$(3-2)-5=-4$
$3-(2-5)=6$ }
≠

N, +	Yes	Z, +	Yes	R, +	Yes
N, ×	Yes	Z, −	No	R, −	No
		Z, ×	Yes	R, ×	Yes

③ $\boxed{\text{Monoid}}$

- Closure
- Associative
- Identity

$$\boxed{a * e = e * a = a}$$

$3+e=3$

$\downarrow$

$e=0 \notin N$

$2 \times e = 2$

$\downarrow$

$e=1 \in N$

N, + NO

N, × yes

Z, + yes

Z, × yes

R, + yes

R, × yes

④ $\boxed{\text{Group}}$

- Closure Property
- Associative Property
- Identity
- Inverse.

$9 \times a^{-1} = e$

$9 \times a^{-1} = 1$

$a = 9$

$\boxed{a^{-1} = \frac{1}{9}} \notin N$

N, × NO

Z, + yes

$-3 + a^{-1} = e$

$-3 + a^{-1} = 0 \quad \therefore a^{-1} = 3 \in Z$

R, + yes

Z, × NO

$-3 \times a^{-1} = e$

$-3 \times a^{-1} = 1 \quad \therefore a^{-1} = \frac{1}{-3} \notin Z$

R, × NO

⑤ $\boxed{\text{Abelian Group}}$

- Closure
- Associative
- Identity
- Inverse
- Commutative.

$$\boxed{a * b = b * a}$$

$-3+2=-1$

$2+(-3)=-1$

Z, + yes

R, + yes.

Q. **Proof that the set Z of all integers with binary operation * defined by a*b = a+b+1 such that ∀ a,b ∈ Z is an abelian group**

$a * b = a+b+1$ ---- given

1) Closure

$a, b \in Z$

$a * b = a+b+1 \in Z$

$\therefore$ satisfy closure property.

2) Associative

$(a * b) * c = (a+b+1) * c$

$= a+b+1+c+1$

$= \boxed{a+b+c+2}$

$a * (b * c)$

$= a * (b+c+1)$

$= a+b+c+1+1$

$= \boxed{a+b+c+2}$

→ Two are same

$\therefore$ Satisfy Associative property

3) Identity : $a * e = a$

$a+e+1 = a$

$\therefore e = -1 \in Z$

$\therefore$ Satisfy Identity.

4> Inverse : $a * a^{-1} = e$

$a + a^{-1} + 1 = e$

$a + a^{-1} + 1 = -1$

$a + a^{-1} = -2$

$\boxed{a^{-1} = -2 - a}$

5> Commutative : $a * b = b * a$

$a + b + 1 = b + a + 1$

same.

$\therefore$ Satisfied.

$\therefore$ $(Z, *)$ is a Abelian Group. (Proved)

Q. If set Q_1 of all rational numbers other than 1 with $a*b = a+b-ab$. SJow that (G,*) is a group.

$a * b = a + b - ab$

1> closure

$a, b \in Q$

$a + b - ab \in Q$

2> Associative

$(a * b) * c = (a + b - ab) * c$

$= (a + b - ab) + c - (a + b - ab) \cdot c$

$= a + b - ab + c - ac - bc + abc$

$= a + b + c - ab - bc - ac + abc$

$a * (b * c) = a * (b + c - bc)$

$= a + (b + c - bc) - a(b + c - bc)$

$= a + b + c - bc - ab - ac + abc$

$\therefore (a * b) * c = a * (b * c)$

So Associative property satisfied.

3> Identity

$a * e = a$

$a + e - ae = a$

$e(1 - a) = 0$

$e = \frac{0}{1-a} = 0 \in Q$

4> Inverse

$a * a^{-1} = e$

$a + a^{-1} - a \cdot a^{-1} = 0$

$a^{-1} - a a^{-1} = -a$

$a^{-1}(1 - a) = -a$

$a^{-1} = -\frac{a}{1-a} \in Q$

$(a \neq 1)$

$\therefore$ Therefore $(G, *)$ is a group.

Q.

t G be the set of all non zero real numbers and let $a * b = \frac{ab}{2}$. Show that $(G, *)$ is an abelian group.

$a * b = \frac{ab}{2}$ - Given.

closure

$a, b \in R,\ a, b \neq 0$

$\frac{ab}{2} \in R,\ ab \neq 0$

$\therefore$ satisfies.

Associative $(a * b) * c = \left(\frac{ab}{2}\right) * c = \frac{abc}{4}$

$a * (b * c) = a * \frac{bc}{2} = \frac{abc}{4}$

$\therefore$ Satisfied associative property.

Identity

$a * e = a$

$\frac{ae}{2} = a$

$e = 2 \in R$

$\& \ 2 \neq 0.$

$\therefore$ Satisfied.

Inverse

$a * a^{-1} = e$

$\frac{aa^{-1}}{2} = 2$

$a^{-1} = \frac{4}{a} \in R$

$\& \ a \neq 0$

$\therefore$ Satisfied.

Commutative

$a * b = b * a$

$\frac{ab}{2} = \frac{ba}{2}$

$\therefore$ Satisfied.

$\therefore (G, *)$ is an Abelian group. (Proved)

Let G={Even,Odd} and binary operation is defined below.Show that (G,*) is an abelian group

*	Even	Odd
Even	Even	Odd
Odd	Odd	Even

① Closure Properties

All the values are part of set G.

② Associative

(Even * Even) * Odd = Even * odd = odd

Even * (Even * Odd) = Even * odd = odd

③ Identity even * e = even

$\boxed{e = \text{even}}$

④ Inverse

$\text{Even} * \text{Even}^{-1} = e$

$\text{Even} * \text{Even}^{-1} = \text{Even}$

$\boxed{\text{Even}^{-1} = \text{Even}}$

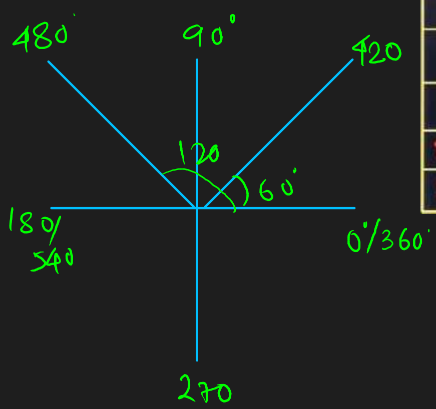
⑤ Commutative

Even * Odd = Odd

Odd * Even = Odd] =

Let R={0 , 60 , 120 , 180°, 240 , 300 } and * is a binary operation so that a and b in R, a*b is overall angular rotation corresponding to successive rotations by a and b. Show that (R,*) is a group

*	0°	60°	120°	180°	240°	300°
0°	0°	60°	120°	180°	240°	300°
60°	60°	120°	180°	240°	300°	0°
120°	120°	180°	240°	300°	0°	60°
180°	180°	240°	300°	0°	60°	120°
240°	240°	300°	0°	60°	120°	180°
300°	300°	0°	60°	120°	180°	240°



Closure

All output $\in R$

Associative

$0 * (60 * 120) = 0 * 180$

$= 180$

$(0 * 60) * 120 = 60 * 120 = 180$

Identity

$a * e = a$

$60 * e = 60$

$\boxed{e = 0}$

Inverse

$a * a^{-1} = e$

$a + a^{-1} = 0$

$60 + a^{-1} = 0 \quad \therefore \boxed{a^{-1} = -60} = 300 \in R.$

Modulo

Addition Modulo :

let,

$$\boxed{\begin{array}{l} a, b \in \mathbb{Z} \\ \wedge \\ m \in \mathbb{Z}^+ \\ m = \text{const.} \end{array}}$$

$\longrightarrow$ Addition modulo is denoted by : $a +_m b$

and defined,

$$\boxed{a +_m b = r \; ; \; 0 \le r \le m}$$

where r is the least non-negative remainder when $(a+b) \% m = 0$.

Q. $5 +_3 9 = 2$ $5+9 = 14 \rightarrow 14 \% 3 = 2$

Q. $15 +_5 25 = 0$

Multiplication Modulo : $a \times_m b$

$$\boxed{a \times_m b = r \; ; \; 0 \le r \le m}$$

Q. $3 \times_4 5 = 3$ Q. $9 \times_4 4 = 0$

Show that (G, $+_8$) is an abelian group where G = (0,1,2,3,4,5,6,7)

$+_8$	0	1	2	3	4	5	6	7
0	0	1	2	3	4	5	6	7
1	1	2	3	4	5	6	7	0
2	2	3	4	5	6	7	0	1
3	3	4	5	6	7	0	1	2
4	4	5	6	7	0	1	2	3
5	5	6	7	0	1	2	3	4
6	6	7	0	1	2	3	4	5
7	7	0	1	2	3	4	5	6

① Closure $\forall a, b \in G$ $a +_8 b \in G$

$\therefore$ Satisfies closure

② Associative $(a +_8 b) +_8 c$ | $a +_8 (b +_8 c)$

$= (1 +_8 2) +_8 3$ | $= 1 +_8 (2 +_8 3)$

$= 3 +_8 3$ | $= 1 +_8 5$

$= 6$ | $= 6$

$\therefore (a +_8 b) +_8 c = a +_8 (b +_8 c)$ $\therefore$ Satisfies associative.

③ Identity : Identity element $= 0$

④ Inverse : $0 +_8 0 = 0$: 0 is the inverse of 0

$1 +_8 7 = 0$: 1 is the inverse of 7

7 " " " 1

⑤ Commutative : $\left.\begin{array}{l} 1 +_8 2 = 3 \\ 2 +_8 1 = 3 \end{array}\right] =$ $\therefore$ Satisfied.

$\therefore (G, +_8)$ is an Abelian group. (Proved)

Prove that the set G= (1,2,3,4,5,6) is a finite Abelian group of order 6 with respect to multiplication modulo 7

1) Closure $\forall a,b \in G \quad a \times_7 b \in G$

2) Associative

$$\left.\begin{array}{l}(1 \times_7 2) \times_7 3 = 2 \times_7 3 = 6 \\ 1 \times_7 (2 \times_7 3) = 1 \times_7 6 = 6\end{array}\right\}$$

3) Identity : identity elmt = 1

4) Inverse : $1 \times_7 1 = 1$

$1 \times_7 2 = 2$

$2 \times_7 4 = 1$

5) Commutative

$2 \times_7 3 = 6$

$3 \times_7 2 = 6$ ✓

$\times_7$	1	2	3	4	5	6
1	1	2	3	4	5	6
2	2	4	6	1	3	5
3	3	6	2	5	1	4
4	4	1	5	2	6	3
5	5	3	1	6	4	2
6	6	5	4	3	2	1

Cyclic Group :

A group G is said to be cyclic if for some a in G, every element x in G can be expressed as a^n, for some integer n.
Thus G is Generated by a i.e. G=(a)

$A \{1, 2, 3, 4, 5, 6\} \times_7$

3 → Generator

$3^1 = 3$ ✓

$3^2 = 3 \times_7 3 = 2$ ✓

$3^3 = 3^2 \times_7 3 = 2 \times_7 3 = 6$ ✓

$3^4 = 3^3 \times_7 3 = 6 \times_7 3 = 4$ ✓

$3^5 = 3^4 \times_7 3 = 4 \times_7 3 = 5$ ✓

$3^6 = 3^5 \times_7 3 = 5 \times_7 3 = 1$ ✓